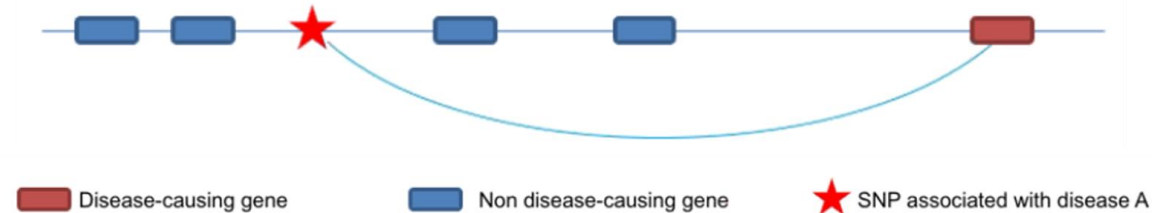


Public bioinformatics resources for GWAS interpretation

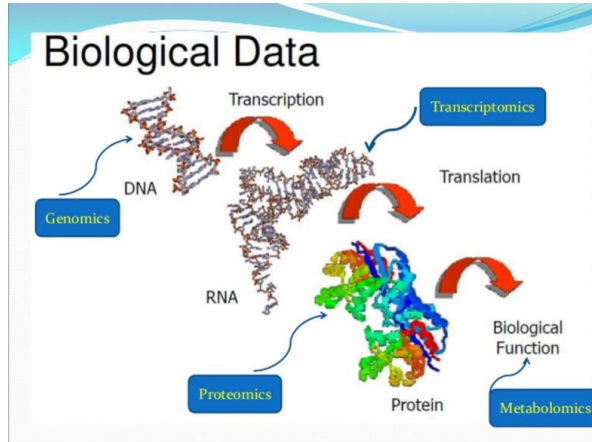
Dr. Konstantinos Hatzikotoulas
Institute of Translational Genomics

Motivation

- Greatest challenge in genetic association studies is to assign functionality (causality) to associated SNPs and genes.
- Predict the most likely genes and variants driving the phenotype associations.
- Need to integrate as much information as possible through genome annotation tools to drive the experimental validation.



What is Bioinformatics?



Computers aid to collect, intergrade and analyse biological data ->
Optimal biological interpretation

Goal of Bioinformatics

- Develop **databases** and **computational tools** to generate knowledge to better understanding a living cell and how it functions at molecular level.



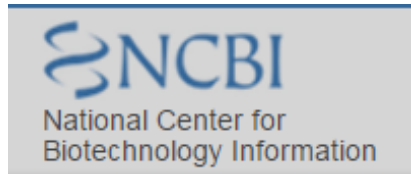
-

- 



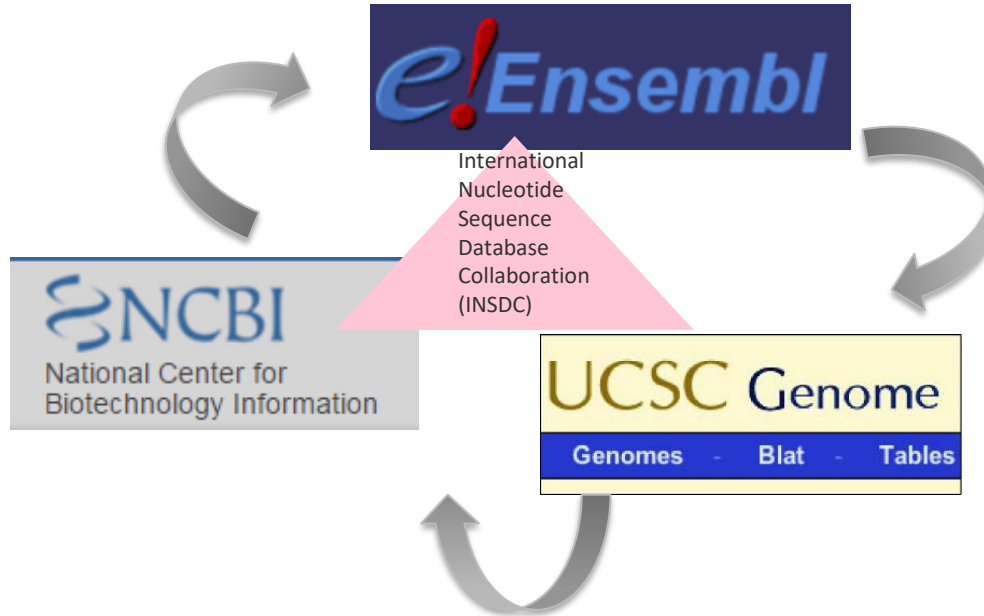
Genome Browsers

- Ensembl : <http://www.ensembl.org>
- UCSC: <http://genome.ucsc.edu/>
- NCBI: <https://www.ncbi.nlm.nih.gov/>



Genome Browsers

- Agreement: To begin bringing all genome assembly data into compliance moving forward



Ensembl ... more than a browser!

- Ensembl is a joint project between the EBI (European Bioinformatics Institute) and the Wellcome Trust Sanger Institute
- **What can I do with Ensembl?**
 - Examine single nucleotide polymorphisms (SNPs) for a gene or chromosomal region.
 - View SNPs across strains (rat, mouse), populations (human) or breeds (dog).
 - View genes with annotations along the chromosome.
 - View alternative transcripts (i.e. splice variants) for a given gene.
 - Upload your own data.
 - Export sequence or create a table of gene information with BioMart.
 - Determine how your variants affect genes and transcripts using the Variant Effect Predictor.
 - Share Ensembl views with your colleagues and collaborators.



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 - Share Ensembl views with your colleagues and collaborators.



Ensembl ... the front page

- It contains lots of information and links to help you navigate Ensembl.

The screenshot shows the Ensembl genome browser front page. Annotations include:

- Link back to homepage**: Points to the Ensembl logo.
- Ensembl tools**: Points to the 'Tools' link in the top navigation bar.
- Blue bar remains visible on every page**: Points to the top navigation bar.
- Search**: Points to the search bar in the top right.
- See the current release number and what's new**: Points to the 'Ensembl Release 106 (Apr 2022)' section.
- Search**: Points to the search bar in the main content area.
- Human GRCh38 p13**: Points to the 'Human GRCh38 p13' link in the 'Favourite genomes' section.
- Mouse GRCh38 p6**: Points to the 'Mouse GRCh38 p6' link in the 'Favourite genomes' section.
- Zebrafish GRCh11**: Points to the 'Zebrafish GRCh11' link in the 'Favourite genomes' section.
- Compare genes across species**: Points to the 'Compare genes across species' link in the bottom navigation bar.
- Find SNPs and other variants for my gene**: Points to the 'Find SNPs and other variants for my gene' link in the bottom navigation bar.
- Gene expression in different tissues**: Points to the 'Gene expression in different tissues' link in the bottom navigation bar.
- Retrieve gene sequence**: Points to the 'Retrieve gene sequence' link in the bottom navigation bar.
- Find a Data Display**: Points to the 'Find a Data Display' link in the bottom navigation bar.
- Use my own data in Ensembl**: Points to the 'Use my own data in Ensembl' link in the bottom navigation bar.

Hint: <https://www.ensembl.org/index.html>

Ensembl ... the front page ... Exercise 1

1. Go to the species homepage for **Giant Panda**. What is the name of the genome assembly for Panda?
2. How long is the Panda genome (in bp)? How many coding genes have been annotated?
3. Repeat for **Human**.



Ensembl ... Exploring variants

➤ Let's have a look at a specific variant, rs143383 (chr20:35438203).

The screenshot displays the Ensembl genome browser interface for variant rs143383. The top navigation bar includes links for BLAST/BLAT, VEP, Tools, BioMart, Downloads, Help & Docs, and Blog. The main header shows the Human (GRCh38.p13) genome assembly. The left sidebar contains a 'Variant displays' section with a tree view for 'Explore this variant', including Genomic context, Genes and regulation, Flanking sequence, Population genetics, Phenotype data, Sample genotypes, Linkage disequilibrium, Phylogenetic context, Citations, and 3D Protein model. Below this are buttons for 'Configure this page', 'Custom tracks', 'Export data', 'Share this page', and 'Bookmark this page'.

The main content area is titled 'rs143383 SNP'. It provides detailed information about the variant, including its most severe consequence (5 prime UTR variant), alleles (G/A), change tolerance (CADD: A:21.4, GERP: 2.76), location (Chromosome 20:35438203), co-located variants (dbSNP rs1555823599), evidence status, clinical significance, HGVS names, synonyms, genotyping chips, original source, and description from SNPedia. A callout box labeled 'Variant information' points to the URL http://www.ensembl.org/info/genome/variation/prediction/predicted_data.html.

Below the variant information is a section titled 'Explore this variant' with a grid of icons representing different data types. A callout box labeled 'Variation icons. These go to the same places as the links on the left' points to this grid. The icons include Genomic context, Genes and regulation, Flanking sequence, Population genetics, Phenotype data, Sample genotypes, Linkage disequilibrium, Phylogenetic context, Citations, and 3D Protein model. Each icon has a number indicating the count of items, such as 2 for Genes and regulation, 61 for Population genetics, 11 for Phenotype data, 3009 for Sample genotypes, and 201 for Citations.

Ensembl ... Exploring variants ...

Variant types

➤ What type of variants is rs143383 (chr20:35438203)?

Ensembl BLAST/BLAT | VEP | Tools | BioMart | Downloads | Help & Docs | Blog

Human (GRCh38.p13) ▾

Location: 20:35,437,703-35,438,703 Variant: rs143383 Jobs ▾

Variant displays

- Explore this variant
- Genomic context
- Genes and regulation
- Flanking sequence
- Population genetics
- Phenotype data
- Sample genotypes
- Linkage disequilibrium
- Phylogenetic context
- Citations
- 3D Protein model

Configure this page

Custom tracks

Export data

Share this page

Bookmark this page

rs143383 SNP

Most severe consequence

Alleles

Change tolerance

Location

Co-located variants

Evidence status ⓘ

Clinical significance ⓘ

HGVS names

Synonyms

Genotyping chips

Original source

About this variant

Description from SNPedia

5 prime UTR variant | See all predicted consequences

GIA | Ancestral: G | Highest population MAF: 0.49

CADD: A:21.4 | GERP: 2.76

Chromosome 20:35438203 (forward strand) | VCF: 20 35438203 rs143383 G A

dbSNP rs1555823599 (G/-) | HGMD-PUBLIC CR072309

AD

This variant has 5 HGVS names - [Show](#)

This variant has 9 synonyms - [Show](#)

This variant has assays on 5 chips - [Show](#)

Variants (including SNPs and indels) imported from dbSNP (release 154) | [View in dbSNP](#)

This variant overlaps 2 transcripts, has 3009 sample genotypes, is associated with 11 phenotypes and is mentioned in 201 citations.

This SNP is associated with [osteoarthritis](#) (OA). It is located in the "five prime untranslated region" (5'UTR) of the gene encoding the embryonic stage onwards. GDF5 is also known as "cartilage-derived morphogenetic protein 1" or "BMP14". ... [Show](#)

Explore this variant ⓘ

- Genomic context
- Genes and regulation
- Flanking sequence
- Population genetics
- Phenotype data
- Sample genotypes
- Linkage disequilibrium
- Phylogenetic context
- Citations
- 3D Protein model

Ensembl ... Exploring variants ...

Variant types

1. Small scale in one or few nucleotides of a gene

Type	Description	Example (Reference / Alternative)
SNP	Single Nucleotide Polymorphism	Ref: ...TTG A CGTA... Alt: ...TTG G CGTA...
Insertion	Insertion of one or several nucleotides	Ref: ...TTGACGTA... Alt: ...TTGAT TC GTA...
Deletion	Deletion of one or several nucleotides	Ref: ...TTG AC GTA... Alt: ...TTGGTA...
Indel	An insertion and a deletion, affecting 2 or more nucleotides	Ref: ...TTG AC GTA... Alt: ...TTG GCT CGTA...
Substitution	A sequence alteration where the length of the change in the variant is the same as that of the reference.	Ref: ...TTG AC GTA... Alt: ...TTG TA GTA...

2. Large scale in chromosomal structure

CNV

Copy Number Variation: increases or decreases the copy number of a given region

Reference:



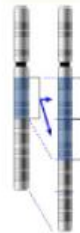
"Gain" of one copy:



"Loss" of one copy:



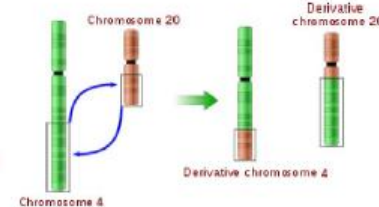
deletion



duplication



insertion

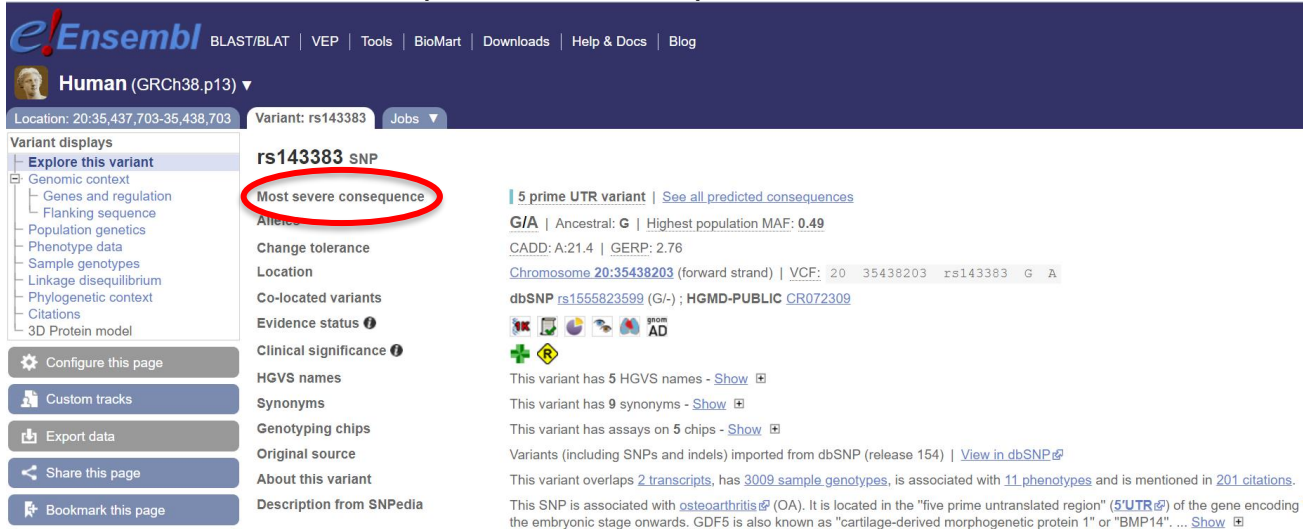


translocation

Ensembl ... Exploring variants ...

Calculated variant consequences

➤ Let's have a look at the predicted consequences of the variant.



e!Ensembl BLAST/BLAT | VEP | Tools | BioMart | Downloads | Help & Docs | Blog

Human (GRCh38.p13) ▾

Location: 20:35,437,703-35,438,703 Variant: rs143383 Jobs ▾

Variant displays

- Explore this variant
- Genomic context
- Genes and regulation
- Flanking sequence
- Population genetics
- Phenotype data
- Sample genotypes
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- Phylogenetic context
- Citations
- 3D Protein model

rs143383 SNP

Most severe consequence

5 prime UTR variant | [See all predicted consequences](#)

GIA | Ancestral: G | Highest population MAF: 0.49
CADD: A:21.4 | GERP: 2.76

Chromosome 20:35438203 (forward strand) | VCF: 20 35438203 rs143383 G A

dbSNP rs1555823599 (G/-) | [HGMD-PUBLIC CR072309](#)

AD

This variant has 5 HGVS names - [Show](#)

This variant has 9 synonyms - [Show](#)

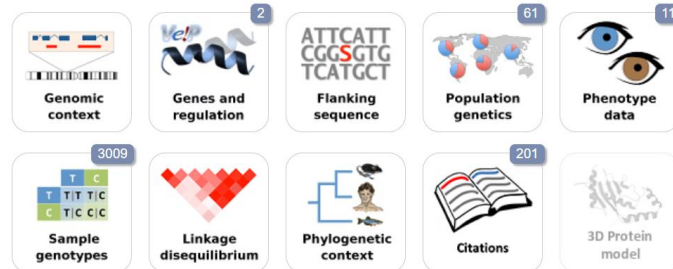
This variant has assays on 5 chips - [Show](#)

Variants (including SNPs and indels) imported from dbSNP (release 154) | [View in dbSNP](#)

This variant overlaps 2 [transcripts](#), has 3009 [sample genotypes](#), is associated with 11 [phenotypes](#) and is mentioned in 201 [citations](#).

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Explore this variant ?



Genomic context 2

Genes and regulation 2

Flanking sequence 61

Population genetics 61

Phenotype data 11

Sample genotypes 3009

Linkage disequilibrium

Phylogenetic context

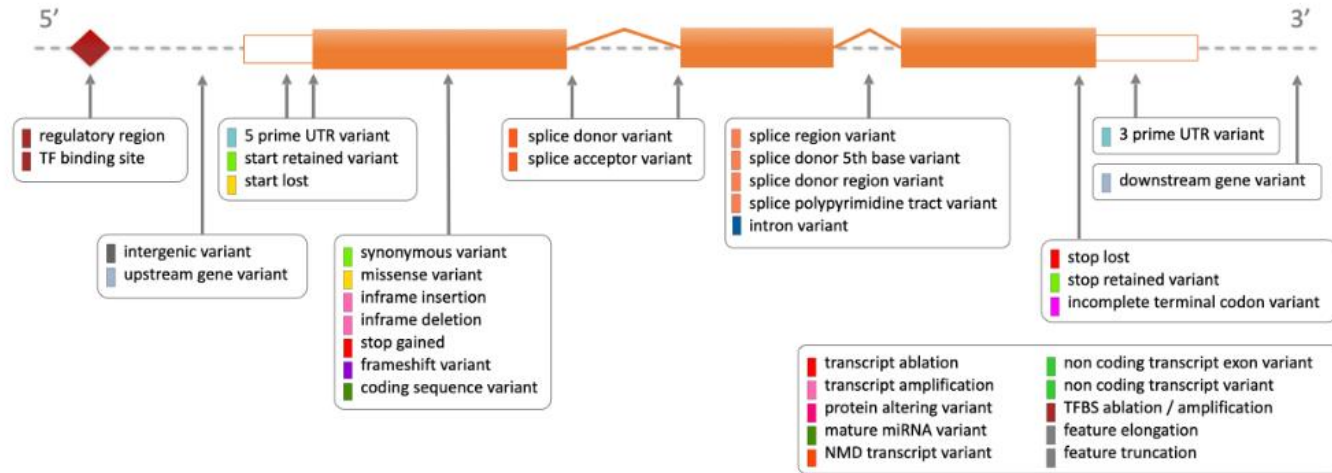
Citations 201

3D Protein model

Ensembl ... Exploring variants ...

Calculated variant consequences

- A rule-based approach to predict the effects that each allele of the variant may have on the transcript.



Ensembl ... Exploring variants ...

Calculated variant consequences

➤ Sequence Ontology (SO) consequence term :

http://www.ensembl.org/info/genome/variation/prediction/predicted_data.html

SO term	SO description	SO accession	Display term	IMPACT
transcript_ablation	A feature ablation whereby the deleted region includes a transcript feature	SO:0001893	Transcript ablation	HIGH
splice_acceptor_variant	A splice variant that changes the 2 base region at the 3' end of an intron	SO:0001574	Splice acceptor variant	HIGH
splice_donor_variant	A splice variant that changes the 2 base region at the 5' end of an intron	SO:0001575	Splice donor variant	HIGH
stop_gained	A sequence variant whereby at least one base of a codon is changed, resulting in a premature stop codon, leading to a shortened transcript	SO:0001587	Stop gained	HIGH
frameshift_variant	A sequence variant which causes a disruption of the translational reading frame, because the number of nucleotides inserted or deleted is not a multiple of three	SO:0001589	Frameshift variant	HIGH
stop_lost	A sequence variant where at least one base of the terminator codon (stop) is changed, resulting in an elongated transcript	SO:0001578	Stop lost	HIGH
start_lost	A codon variant that changes at least one base of the canonical start codon	SO:0002012	Start lost	HIGH
transcript_amplification	A feature amplification of a region containing a transcript	SO:0001889	Transcript amplification	HIGH
inframe_insertion	An inframe non synonymous variant that inserts bases into the coding sequence	SO:0001821	Inframe insertion	MODERATE
inframe_deletion	An inframe non synonymous variant that deletes bases from the coding sequence	SO:0001822	Inframe deletion	MODERATE
missense_variant	A sequence variant, that changes one or more bases, resulting in a different amino acid sequence but where the length is preserved	SO:0001583	Missense variant	MODERATE
protein_altering_variant	A sequence variant which is predicted to change the protein encoded in the coding sequence	SO:0001818	Protein altering variant	MODERATE
splice_region_variant	A sequence variant in which a change has occurred within the region of the splice site, either within 1-3 bases of the exon or 3-8 bases of the intron	SO:0001630	Splice region variant	LOW
splice_donor_5th_base_variant	A sequence variant that causes a change at the 5th base pair after the start of the intron in the orientation of the transcript	SO:0001787	Splice donor 5th base variant	LOW
splice_donor_region_variant	A sequence variant that falls in the region between the 3rd and 6th base after splice junction (5' end of intron)	SO:0002170	Splice donor region variant	LOW
splice_polypyrimidine_tract_variant	A sequence variant that falls in the polypyrimidine tract at 3' end of intron between 17 and 3 bases from the end (acceptor -3 to acceptor -17)	SO:0002169	Splice polypyrimidine tract variant	LOW
incomplete_terminal_codon_variant	A sequence variant where at least one base of the final codon of an incompletely annotated transcript is changed	SO:0001626	Incomplete terminal codon variant	LOW
start_retained_variant	A sequence variant where at least one base in the start codon is changed, but the start remains	SO:0002013	Start retained variant	LOW
stop_retained_variant	A sequence variant where at least one base in the terminator codon is changed, but the terminator remains	SO:0001567	Stop retained variant	LOW
synonymous_variant	A sequence variant where there is no resulting change to the encoded amino acid	SO:0001819	Synonymous variant	LOW
coding_sequence_variant	A sequence variant that changes the coding sequence	SO:0001580	Coding sequence variant	MODIFIER
mature_miRNA_variant	A transcript variant located with the sequence of the mature miRNA	SO:0001620	Mature miRNA variant	MODIFIER
5_prime_UTR_variant	A UTR variant of the 5' UTR	SO:0001623	5 prime UTR variant	MODIFIER
3_prime_UTR_variant	A UTR variant of the 3' UTR	SO:0001624	3 prime UTR variant	MODIFIER
non_coding_transcript_exon_variant	A sequence variant that changes non-coding exon sequence in a non-coding transcript	SO:0001792	Non coding transcript exon variant	MODIFIER
intron_variant	A transcript variant occurring within an intron	SO:0001627	Intron variant	MODIFIER
NMD_transcript_variant	A variant in a transcript that is the target of NMD	SO:0001621	NMD transcript variant	MODIFIER
non_coding_transcript_variant	A transcript variant of a non coding RNA gene	SO:0001619	Non coding transcript variant	MODIFIER
upstream_gene_variant	A sequence variant located 5' of a gene	SO:0001631	Upstream gene variant	MODIFIER
downstream_gene_variant	A sequence variant located 3' of a gene	SO:0001632	Downstream gene variant	MODIFIER
TFBS_ablation	A feature ablation whereby the deleted region includes a transcription factor binding site	SO:0001895	TFBS ablation	MODIFIER
TFBS_amplification	A feature amplification of a region containing a transcription factor binding site	SO:0001892	TFBS amplification	MODIFIER
TF_binding_site_variant	A sequence variant located within a transcription factor binding site	SO:0001782	TF binding site variant	MODIFIER
regulatory_region_ablation	A feature ablation whereby the deleted region includes a regulatory region	SO:0001894	Regulatory region ablation	MODERATE
regulatory_region_amplification	A feature amplification of a region containing a regulatory region	SO:0001891	Regulatory region amplification	MODIFIER
feature_elongation	A sequence variant that causes the extension of a genomic feature, with regard to the reference sequence	SO:0001907	Feature elongation	MODIFIER
regulatory_region_variant	A sequence variant located within a regulatory region	SO:0001566	Regulatory region variant	MODIFIER
feature_truncation	A sequence variant that causes the reduction of a genomic feature, with regard to the reference sequence	SO:0001906	Feature truncation	MODIFIER
intergenic_variant	A sequence variant located in the intergenic region, between genes	SO:0001628	Intergenic variant	MODIFIER

Ensembl ... Exploring variants ...

Calculated variant consequences

➤ Let's have a look at all the predicted consequences of the variant.

The screenshot displays the Ensembl genome browser interface for variant rs143383. The top navigation bar includes links for BLAST/BLAT, VEP, Tools, BioMart, Downloads, Help & Docs, and Blog. The main header shows the Human (GRCh38.p13) genome and the variant location (20:35,437,703-35,438,703). The variant is identified as rs143383 SNP.

Variant displays

- Explore this variant
- Genomic context
- Genes and regulation
- Flanking sequence
- Population genetics
- Phenotype data
- Sample genotypes
- Linkage disequilibrium
- Phylogenetic context
- Citations
- 3D Protein model

Variant details:

- rs143383 SNP**
- Most severe consequence:** 5 prime UTR variant (circled in red)
- Alleles:** G/A | Ancestral: G | Highest population frequency: 0.49
- Change tolerance:** CADD: A:21.4 | GERP: 2.76
- Location:** Chromosome 20:35438203 (forward strand) | VCF: 20 35438203 rs143383 G A
- Co-located variants:** dbSNP rs1555823599 (G/-) | HGMD-PUBLIC CR072309
- Evidence status:** (Icons for various evidence types)
- Clinical significance:** (Icon for clinical significance)
- HGVS names:** This variant has 5 HGVS names - [Show](#)
- Synonyms:** This variant has 9 synonyms - [Show](#)
- Genotyping chips:** This variant has assays on 5 chips - [Show](#)
- Original source:** Variants (including SNPs and indels) imported from dbSNP (release 154) | [View in dbSNP](#)
- About this variant:** This variant overlaps 2 [transcripts](#), has 3009 [sample genotypes](#), is associated with 11 [phenotypes](#) and is mentioned in 201 [citations](#). This SNP is associated with [osteoarthritis](#) (OA). It is located in the "five prime untranslated region" (5'UTR) of the gene encoding the embryonic stage onwards. GDF5 is also known as "cartilage-derived morphogenetic protein 1" or "BMP14". ... [Show](#)
- Description from SNPedia:**

Explore this variant

The "Explore this variant" section features a grid of interactive icons, each with a count of associated data points:

- Genomic context (2)
- Genes and regulation (2)
- Flanking sequence (61)
- Population genetics (11)
- Phenotype data (11)
- Sample genotypes (3009)
- Linkage disequilibrium
- Phylogenetic context
- Citations (201)
- 3D Protein model

Ensembl ... Exploring variants ...

Calculated variant consequences

➤ This variant is found in 2 transcripts of the *GDF5* gene.

Genes and regulation

Gene and Transcript consequences



Gene	Transcript (strand)	Allele (Tr. allele)	Consequence Type	Position in transcript	Position in CDS	Position in protein	AA	Codons	Detail
ENSG00000125965	ENST00000374369.8 (-)	A	5 prime UTR variant	26 (out of 2368)	-	-	-	-	Show
HGNC: GDF5	biotype: protein_coding	(T)							
ENSG00000125965	ENST00000374372.1 (-)	A	Intron variant	-	-	-	-	-	Show
HGNC: GDF5	biotype: protein_coding	(T)							



coding_sequence_variant	A sequence variant that changes the coding sequence	SO:0001580	Coding sequence variant	MODIFIER
mature_miRNA_variant	A transcript variant located with the sequence of the mature miRNA	SO:0001620	Mature miRNA variant	MODIFIER
5_prime_UTR_variant	A UTR variant of the 5' UTR	SO:0001623	5 prime UTR variant	MODIFIER
3_prime_UTR_variant	A UTR variant of the 3' UTR	SO:0001624	3 prime UTR variant	MODIFIER
non_coding_transcript_exon_variant	A sequence variant that changes non-coding exon sequence in a non-coding transcript	SO:0001792	Non coding transcript exon variant	MODIFIER
intron_variant	A transcript variant occurring within an intron	SO:0001627	Intron variant	MODIFIER
NMD_transcript_variant	A variant in a transcript that is the target of NMD	SO:0001621	NMD transcript variant	MODIFIER
non_coding_transcript_variant	A transcript variant of a non coding RNA gene	SO:0001619	Non coding transcript variant	MODIFIER



Variant: rs143383 Jobs

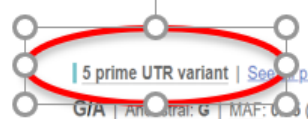
rs143383 SNP

Most severe consequence
Alleles
Change tolerance

5 prime UTR variant | See predicted consequences

G/A | A: 0.49 | A: 0.49 | MAF: 0.49 (A) | Highest population MAF: 0.49

CADD: A:20.6 | GERP: 2.34



Ensembl ... Exploring variants...

Population genetics

➤ Let's have a look at population genetics.

e!Ensembl BLAST/BLAT | VEP | Tools | BioMart | Downloads | Help & Docs | Blog

Human (GRCh38.p13) ▾

Location: 20:35,437,703-35,438,703 Variant: rs143383 Jobs ▾

Variant displays

- Explore this variant
- Genomic context
- Genes and regulation
- Flanking sequence
- Population genetics**
- Phenotype data
- Sample genotypes
- Linkage disequilibrium
- Phylogenetic context
- Citations
- 3D Protein model

Configure this page

Custom tracks

Export data

Share this page

Bookmark this page

rs143383 SNP

Most severe consequence

Alleles

Change tolerance

Location

Co-located variants

Evidence status ⓘ

Clinical significance ⓘ

HGVs names

Synonyms

Genotyping chips

Original source

About this variant

Description from SNPedia

5 prime UTR variant | [See all predicted consequences](#)

GIA | Ancestral: **G** | Highest population MAF: **0.49**

CADD: A:21.4 | GERP: 2.76

Chromosome **20:35438203** (forward strand) | VCF: 20 35438203 rs143383 G A

SNP rs155582355 (14) : **HGMD-PUBLIC CR072309**

This variant has 5 HGVS names - [Show](#) ⓘ

This variant has 9 synonyms - [Show](#) ⓘ

This variant has assays on 5 chips - [Show](#) ⓘ

Variants (including SNPs and indels) imported from dbSNP (release 154) | [View in dbSNP](#) ⓘ

This variant overlaps 2 [transcripts](#), has [3009 sample genotypes](#), is associated with [11 phenotypes](#) and is mentioned in [201 citations](#).

This SNP is associated with [osteoarthritis](#) (OA). It is located in the "five prime untranslated region" (**5'UTR** ⓘ) of the gene encoding the embryonic stage onwards. GDF5 is also known as "cartilage-derived morphogenetic protein 1" or "BMP14". ... [Show](#) ⓘ

Explore this variant ⓘ

- Genomic context
- Genes and regulation
- Flanking sequence
- Population genetics**
- Phenotype data
- Sample genotypes
- Linkage disequilibrium
- Phylogenetic context
- Citations
- 3D Protein model

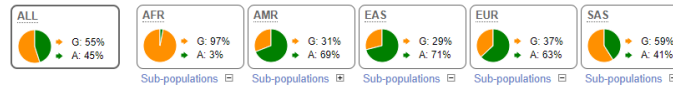
Ensembl ... Exploring variants ...

Population genetics

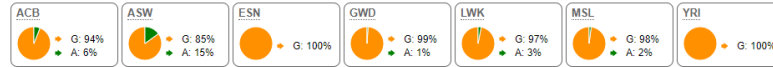
- Frequencies are grouped by project for studies with multiple subpopulations.
- Pie charts can be displayed only for 1000 Genomes allele frequencies.

Population genetics

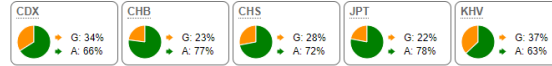
1000 Genomes Project Phase 3 allele frequencies



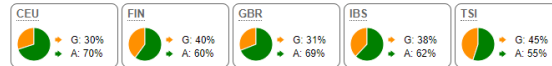
AFR sub-populations



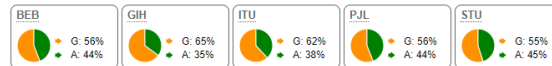
EAS sub-populations



EUR sub-populations



SAS sub-populations



Tables of more detailed data

EAS	G: 0.287 (289)	A: 0.713 (719)	GI: 0.085 (43)	AJ: 0.512 (298)	AI: 0.403 (203)
CDX	G: 0.344 (64)	A: 0.656 (122)	GI: 0.140 (13)	AJ: 0.452 (42)	AI: 0.409 (38)
CHB	G: 0.233 (48)	A: 0.767 (158)	GI: 0.049 (5)	AJ: 0.583 (60)	AI: 0.369 (38)
CHS	G: 0.276 (68)	A: 0.724 (152)	GI: 0.067 (7)	AJ: 0.514 (54)	AI: 0.419 (44)
JPT	G: 0.221 (46)	A: 0.779 (162)	GI: 0.067 (7)	AJ: 0.625 (65)	AI: 0.308 (32)
KHV	G: 0.369 (73)	A: 0.631 (125)	GI: 0.111 (11)	AJ: 0.374 (37)	AI: 0.515 (51)

gnomAD genomes v3.1.2 (11)

Population	Allele: frequency (count)
gnomADg:ALL	G: 0.522 (79301) A: 0.477 (72469)
gnomADg:afri	G: 0.879 (36292) A: 0.121 (5006)
gnomADg:ami	G: 0.401 (365) A: 0.599 (545)
gnomADg:amr	G: 0.376 (5729) A: 0.624 (9491)
gnomADg:asj	G: 0.455 (1576) A: 0.545 (1890)
gnomADg:eas	G: 0.283 (1462) A: 0.717 (3710)
gnomADg:fin	G: 0.415 (4371) A: 0.585 (6163)
gnomADg:mid	G: 0.494 (156) A: 0.506 (160)
gnomADg:nfe	G: 0.377 (25640) A: 0.623 (42306)
gnomADg:oth	G: 0.485 (1010) A: 0.515 (1072)
gnomADg:sas	G: 0.559 (2700) A: 0.441 (2126)

Jump to: [1000 Genomes Project Phase 3 \(32\)](#) | [gnomAD genomes v3.1.2 \(11\)](#) | [NCBI ALFA \(12\)](#) | [TOPMed \(1\)](#) | [Gambian Genome Variation Project \(5\)](#)

Ensembl ... Exploring variants...

Phenotype/Disease data

➤ Let's have a look at the phenotypes associated with this variant.

e!Ensembl BLAST/BLAT | VEP | Tools | BioMart | Downloads | Help & Docs | Blog

Human (GRCh38.p13) ▾

Location: 20:35,437,703-35,438,703 Variant: rs143383 Jobs ▾

Variant displays

- Explore this variant
- Genomic context
- Genes and regulation
- Flanking sequence
- Population genetics
- Phenotype data**
- Sample genotypes
- Linkage disequilibrium
- Phylogenetic context
- Citations
- 3D Protein model

Configure this page Custom tracks Export data Share this page Bookmark this page

rs143383 SNP

Most severe consequence
Alleles
Change tolerance
Location
Co-located variants
Evidence status ⓘ
Clinical significance ⓘ
HGVS names
Synonyms
Genotyping chips
Original source
About this variant
Description from SNPedia

5 prime UTR variant | [See all predicted consequences](#)

GIA | Ancestral: G | Highest population MAF: 0.49
CADD: A:21.4 | GERP: 2.76

Chromosome 20:35438203 (forward strand) | VCF: 20 35438203 rs143383 G A

dbSNP rs1555823599 (G/-) | [HGMD-PUBLIC CR072309](#)

This variant has 5 HGVS names - [Show](#) ⓘ
This variant has 9 synonyms - [Show](#) ⓘ
This variant has assays on 5 chips - [Show](#) ⓘ
Variants (including SNPs and indels) imported from dbSNP (release 154) | [View in dbSNP](#) ⓘ
This variant overlaps 2 [transcripts](#), has 3009 [sample genotypes](#), is associated with 11 [phenotypes](#) and is mentioned in 201 [citations](#).
This SNP is associated with [osteoarthritis](#) (OA). It is located in the "five prime untranslated region" ([5'UTR](#) ⓘ) of the gene encoding the embryonic stage onwards. GDF5 is also known as "cartilage-derived morphogenetic protein 1" or "BMP14". ... [Show](#) ⓘ

Explore this variant ⓘ

Genomic context

Genes and regulation 2

Flanking sequence

Population genetics 61

Phenotype data 19

Sample genotypes 3009

Linkage disequilibrium

Phylogenetic context

Citations 201

3D Protein model

Ensembl ... Exploring variants...

Phenotype/Disease data

➤ This variant is associated with 19 phenotypes/diseases.

Phenotype Data

Significant association(s)

Show All entries	Show/hide columns																			
Phenotype, disease and trait	Source(s)	Mapped Terms	Ontology Accessions	Supporting evidence	External reference	Clinical significance	Reported gene(s)	Associated allele	Statistics											
ACROMESOMELIC DYSPLASIA 2B	ClinVar [Illumina Laboratory ...]	-	Orphanet.93437 #	-	-	★ ★ ★ ★ ★	LOC109461476, GDF5	-												
Acromesomelic dysplasia 2C, Hunter-Thompson type	ClinVar [Illumina Laboratory ...]	-	-	-	-	★ ★ ★ ★ ★	LOC109461476, GDF5	-												
Brachydactyly	ClinVar [Illumina Laboratory ...]	Brachydactyly	HP.0001156 # , Orphanet.294937 #	-	-	★ ★ ★ ★ ★	LOC109461476, GDF5	-												
ClinVar: phenotype not specified	ClinVar [Breakthrough Genomic...]	-	-	-	-	★ ★ ★ ★ ★	LOC109461476, GDF5	-												
Cortical surface area	NHGRI/EBI GWAS catalog #	-	EEO.0010736 #	-	PMID.34560273 #	-	-	G	p-value: 1.00e-17											
Grebe syndrome	ClinVar [Illumina Laboratory ...]	-	Orphanet.2098 #	-	-	★ ★ ★ ★ ★	LOC109461476, GDF5	-												
Height	NHGRI/EBI GWAS catalog #	-	EEO.0004339 #	-	PMID.31562340 #	-	GDF5	-	p-value: 9.00e-78 beta coefficient: 0.06367727 unit increase											
Height	NHGRI/EBI GWAS catalog #	-	EEO.0004339 #	-	PMID.34270706 #	-	GDF5	-	p-value: 1.00e-9 beta coefficient: 0.48 unit increase											
Height	GIANT #	-	EEO.0004339 #	-	-	-	-	G	p-value: 2.81e-35											
Idiopathic knee osteoarthritis	NHGRI/EBI GWAS catalog #	osteoarthritis, knee	EEO.0004616 #	-	PMID.37247657 #	-	-	G	p-value: 2.00e-12 odds ratio: 0.96											
Knee osteoarthritis	NHGRI/EBI GWAS catalog #	Knee osteoarthritis, osteoarthritis, knee	EEO.0004616 # , HP.0005086 #	-	PMID.30374099 #	-	-	I	p-value: 8.00e-18 odds ratio: 1.09287											
Knee osteoarthritis	NHGRI/EBI GWAS catalog #	Knee osteoarthritis, osteoarthritis, knee	EEO.0004616 # , HP.0005086 #	-	PMID.30664745 #	-	GDF5	-	p-value: 5.00e-7 odds ratio: 1.0495905											
MULTIPLE SYNOSTOSES SYNDROME 2	ClinVar [Illumina Laboratory ...]	-	Orphanet.3237 #	-	-	★ ★ ★ ★ ★	LOC109461476, GDF5	-												
OSTEOARTHRITIS SUSCEPTIBILITY 5	ClinVar [OMIM]	Osteoarthritis	EEO.0002506 # , HP.0002758 #	PMID.17384641 #	MIM.601146 #	★ ★ ★ ★ ★	LOC109461476, GDF5	-												
Prostate Cancer	NHGRI/EBI GWAS catalog #	prostate adenocarcinoma, Prostate cancer, prostate carcinoma	EEO.0000673 # , EEO.0001663 # , HP.0012125 # , Orphanet.1331 #	-	PMID.37945903 #	-	-	A	p-value: 1.00e-12 odds ratio: 1.04											
Prostate Cancer	NHGRI/EBI GWAS catalog #	prostate adenocarcinoma, Prostate cancer, prostate carcinoma	EEO.0000673 # , EEO.0001663 # , HP.0012125 # , Orphanet.1331 #	-	PMID.37945903 #	-	-	A	p-value: 3.00e-10 odds ratio: 1.04											
T2 brain MRIs Unsupervised Deep learning derived Imaging Phenotypes dimension 93	NHGRI/EBI GWAS catalog #	-	EEO.0004464 #	-	PMID.38580639 #	-	-	-	p-value: 1.00e-23											
Vertex-wise cortical surface area	NHGRI/EBI GWAS catalog #	-	EEO.0010736 #	-	PMID.34910505 #	-	-	G	p-value: 2.00e-16 beta coefficient: 8.25 z score increase											
Whole brain free water diffusion multivariate analysis	NHGRI/EBI GWAS catalog #	-	EEO.0004464 #	-	PMID.35505052 #	-	-	-	p-value: 3.00e-15											

Ensembl ... Exploring variants...

Citations

➤ Let's have a look at the citations of the variant.

The screenshot shows the Ensembl genome browser interface for variant rs143383. The top navigation bar includes links for BLAST/BLAT, VEP, Tools, BioMart, Downloads, Help & Docs, and Blog. The main header indicates the variant is in the Human (GRCh38.p13) assembly, located at 20:35,437,703-35,438,703. The left sidebar, titled 'Variant displays', lists various tracks: Genomic context, Genes and regulation, Flanking sequence, Population genetics, Phenotype data, Sample genotypes, Linkage disequilibrium, Phylogenetic context, and Citations (which is circled in red). Below this sidebar is a 'Variant displays' section with options like 'Explore this variant', 'Configure this page', 'Custom tracks', 'Export data', 'Share this page', and 'Bookmark this page'. The main content area displays details for variant rs143383, including its most severe consequence (GIA), ancestral allele (G), and population MAF (0.49). It also shows the variant's location on chromosome 20 and its association with dbSNP rs1555823599. The 'Citations' section is highlighted with a red circle, showing a list of 201 citations. The bottom section, 'Explore this variant', features a grid of tiles for different genomic contexts: Genomic context (2), Genes and regulation (2), Flanking sequence (61), Population genetics (61), Phenotype data (11), Sample genotypes (3009), Linkage disequilibrium, Phylogenetic context, Citations (201, circled in red), and 3D Protein model.

Human (GRCh38.p13) ▼

Location: 20:35,437,703-35,438,703 Variant: rs143383 Jobs ▼

Variant displays

- Explore this variant
- Genomic context
- Genes and regulation
- Flanking sequence
- Population genetics
- Phenotype data
- Sample genotypes
- Linkage disequilibrium
- Phylogenetic context
- Citations
- 3D Protein model

Configure this page

Custom tracks

Export data

Share this page

Bookmark this page

rs143383 SNP

Most severe consequence: **GIA** | Ancestral: **G** | Highest population MAF: **0.49**

CADD: A:21.4 | GERP: 2.76

Location: **Chromosome 20:35438203** (forward strand) | VCF: 20 35438203 rs143383 G A

Co-located variants: **dbSNP:rs1555823599 (G/-) ; HGMD-PUBLIC CR072309**

Evidence status:

Clinical significance:

HGVS names: This variant has 5 HGVS names - [Show](#)

Synonyms: This variant has 9 synonyms - [Show](#)

Genotyping chips: This variant has assays on 5 chips - [Show](#)

Original source: Variants (including SNPs and indels) imported from dbSNP (release 154) | [View in dbSNP](#)

About this variant: This variant overlaps 2 [transcripts](#), has [3009 sample genotypes](#), is associated with [11 phenotypes](#) and is mentioned in [201 citations](#).

Description from SNPedia: This SNP is associated with [osteoarthritis](#) (OA). It is located in the "five prime untranslated region" ([5'UTR](#)) of the gene encoding the embryonic stage onwards. GDF5 is also known as "cartilage-derived morphogenetic protein 1" or "BMP14". ... [Show](#)

Explore this variant

- Genomic context (2)
- Genes and regulation (2)
- Flanking sequence (61)
- Population genetics (61)
- Phenotype data (11)
- Sample genotypes (3009)
- Linkage disequilibrium
- Phylogenetic context
- Citations (201)
- 3D Protein model

Ensembl ... Exploring variants ... Citations

➤ This variant is mentioned in 253 citations.

Citations

rs143383 is mentioned in the following publications

Show	All	entries	Show/hide columns	Filter		
Year	PMID	Phenotype	Title	Author(s)	Full text	Citation source
2021	33235627		Identification of key genes in osteoarthritis using bioinformatics, principal component analysis and meta-analysis.	Sun X, Duan H, Xiao L, Yao S, et al	PMC7678638	EPMC
2021	34206824		Current Evidence about Developmental Dysplasia of the Hip in Pregnancy.	Simionescu AA, Cirstoiu MM, Cirstoiu C, Stanescu AMA, et al	PMC8305660	EPMC
2021	34210342		The effect of common variants in GDF5 gene on the susceptibility to chronic postsurgical pain.	Yan S, Nie H, Bu G, Yuan W, et al	PMC8247225	EPMC
2021	33817978		The Mechanisms and Functions of GDF-5 in Intervertebral Disc Degeneration.	Guo S, Cui L, Xiao C, Wang C, et al	PMC8126946	EPMC
2021	33522652		Growth differentiation factor 5 in cartilage and osteoarthritis: A possible therapeutic candidate.	Sun K, Guo J, Yao X, Guo Z, et al	PMC7941218	EPMC
2021	34203285		Genetic Study of IL6, GDF5 and PAPP2 in Association with Developmental Dysplasia of the Hip.	Harsanyi S, Zamborsky R, Krajcova L, Kokavec M, et al	PMC8303839	EPMC
2021	33973562		A Case-Control Study of Major genetic pre-disposition risk alleles in developing DDD in the Northeast US population: Effects of Gene-Gene Interactions.	Valtchinov VI, Zhai BK, Hida T, Lacson R, et al		EPMC
2021	33861510		Frequency of Growth Differentiation Factor 5 rs143383 and asporin D-repeat polymorphisms in patients with hand and knee osteoarthritis in Kurdistan province, Iran.	Moghimi N, Nasser S, Ghafouri F, Jalili A.		EPMC
2020	31923126		A Meta-analysis Assessing the Association Between COL11A1 and GDF5 Genetic Variants and Intervertebral Disc Degeneration Susceptibility.	Wu F, Huang X, Zhang Z, Shao Z.		dbSNP
2020	32103374		A novel variant near LSP1P3 is associated with knee osteoarthritis in the Chinese population.	Li Y, Liu F, Xu X, Zhang H, et al		EPMC,dbSNP
2020	31932746		Regulation of Gdf5 expression in joint remodelling, repair and osteoarthritis.	Kania K, Colella F, Riemen AHK, Wang H, et al	PMC6957535	EPMC
2020	32928309		Association between growth differentiation factor 5 rs143383 genetic polymorphism and the risk of knee osteoarthritis among Caucasian but not Asian: a meta-analysis.	Peng L, Jin S, Lu J, Ouyang C, et al	PMC7488690	EPMC
2020	32244273		Developmental Dysplasia of the Hip: A Review of Etiopathogenesis, Risk Factors, and Genetic Aspects.	Harsanyi S, Zamborsky R, Krajcova L, Kokavec M, et al	PMC7230892	EPMC
2020	32967120		Can Implementation of Genetics and Pharmacogenomics Improve Treatment of Chronic Low Back Pain?	Suntsov V, Jovanovic F, Knezevic E, Candido KD, et al	PMC7558486	EPMC
2020	32144293		Generation and characterization of human induced pluripotent stem cells (iPSCs) from hand osteoarthritis patient-derived fibroblasts.	Castro-Vila R, Sanjurjo-Rodríguez C, Piñeiro-Ramil M, Lomida-Ramos T, et al	PMC7060311	EPMC

Ensembl ... Exploring genes

➤ Let's have a look at *GDF5* (growth differentiation factor 5).

Ensembl BLAST/BLAT | VEP | Tools | BioMart | Downloads | Help & Docs | Blog

Human (GRCh38.p14) | Location: 20:35,433,347-35,454,740 | Gene: GDF5 | Variant: rs143383

Gene: GDF5 ENSG00000125905

Description growth differentiation factor 5 [Source: HGNC Symbol; Acc: HGNC:42204]

Gene Synonyms BMP14, CDMP1

Location Chromosome 20: 35,433,347-35,454,740 reverse strand. GRCh38: CM000682.2

About this gene This gene has 3 transcripts (splice variants), 193 orthologues, 31 paralogues and is associated with 20 phenotypes.

Transcripts [Show transcript table](#)

Variant table

This table shows known variants for this gene. Use the 'Consequence Type' filter to view a subset of these.

Filter:

Variant ID	Chr. bp	Alleles	Global MAF	Class	Source	Evidence	Clin. Sig.	Conseq. Type	AA	AA coord	SIFT	PolyPhen	CADD	REVEL	MetaLR	Mutation Assessor	Transcript
rs93400324	20:35433342-35433382	CACAGTT * - TTAGGCA CAGT /C ACAGTTT T	-	indel	dbSNP	AD	-	3 prime UTR variant	-	-	-	-	-	-	-	-	ENST00000374369.8
rs2062450959	20:35433353	G/T	-	SNP	dbSNP	-	-	3 prime UTR variant	-	-	-	-	-	-	-	-	ENST00000374369.8
rs1392834028	20:35433354-35433357	CACA/CA	-	indel	dbSNP	AD	-	3 prime UTR variant	-	-	-	-	-	-	-	-	ENST00000374369.8
rs2062450990	20:35433356	C/G	-	SNP	dbSNP	-	-	3 prime UTR variant	-	-	-	-	-	-	-	-	ENST00000374369.8
rs2146577401	20:35433358	G/A	-	SNP	dbSNP	-	-	3 prime UTR variant	-	-	-	-	-	-	-	-	ENST00000374369.8
rs2062451012	20:35433360	T/G	-	SNP	dbSNP	AD	-	3 prime UTR variant	-	-	-	-	-	-	-	-	ENST00000374369.8
rs972378008	20:35433362	T/C	-	SNP	dbSNP	AD	-	3 prime UTR variant	-	-	-	-	-	-	-	-	ENST00000374369.8
rs1297006726	20:35433364	C/T	-	SNP	dbSNP	-	-	3 prime UTR variant	-	-	-	-	-	-	-	-	ENST00000374369.8
rs1466129611	20:35433365	T/A/G	-	SNP	dbSNP	AD	-	3 prime UTR variant	-	-	-	-	-	-	-	-	ENST00000374369.8
rs1466129611	20:35433365	T/A/G	-	SNP	dbSNP	AD	-	3 prime UTR variant	-	-	-	-	-	-	-	-	ENST00000374369.8
rs2062451074	20:35433369	T/C	-	SNP	dbSNP	-	-	3 prime UTR variant	-	-	-	-	-	-	-	-	ENST00000374369.8
rs548379983	20:35433370	A/G	< 0.001 (0)	SNP	dbSNP	AD	-	3 prime UTR variant	-	-	-	-	-	-	-	-	ENST00000374369.8

Ensembl ... Exploring variants ... Exercise 2

Explore rs2075650 (build 37).

1. What type of variants is?
2. Where is this variant located (chr - position - strand)?
3. Are there big differences in allele frequencies between populations?
4. What is the ancestral allele frequency in the Japanese in Tokyo (JPT) population from the 1000G set?
5. What is the least frequent genotype for this variant in the KHV population from the 1000G set?
6. In how many transcripts is this variant found?
7. Are all transcripts from the same gene?
8. How many and which consequence type are predicted for this variant?
9. How many phenotypes are associated with this variant?
10. How many phenotypes are associated with the gene that this variant is located in?
11. How many publications refer to rs2075650?



- 



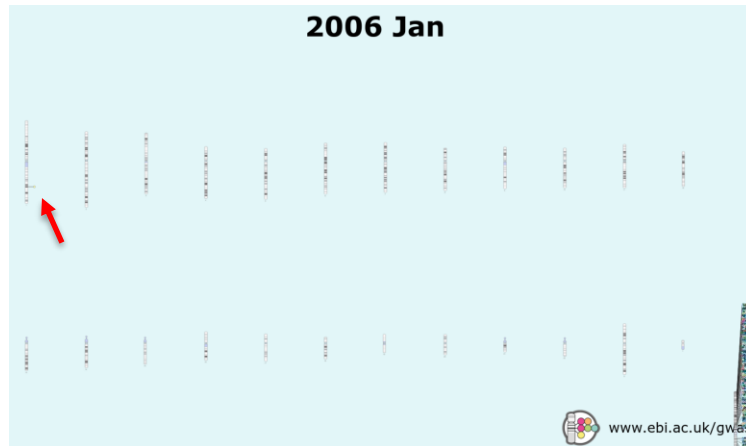
Human Traits & Diseases Databases

- GWAS catalog (<https://www.ebi.ac.uk/gwas/>)
- OMIM (<http://omim.org/>)
- Orphanet (<http://www.orpha.net>)



Human Traits & Diseases Databases GWAS catalog

- “GWAS catalog” is a catalog of published/unpublished genome-wide association studies.



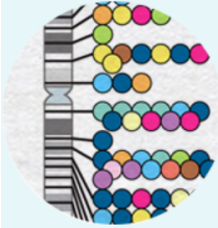
2022



Show SNPs for		
	Digestive system disease	640
	Cardiovascular disease	1607
	Metabolic disease	1896
	Immune system disease	2380
	Nervous system disease	3120
	Liver enzyme measurement	1363
	Lipid or lipoprotein measurement	4772
	Inflammatory marker measurement	469
	Hematological measurement	11555
	Body weights and measures	3984
	Cardiovascular measurement	1875
	Other measurement	27644
	Response to drug	446
	Biological process	2523
	Cancer	2713
	Other disease	3805
	Other trait	2822

Human Traits & Diseases Databases

GWAS catalog



GWAS Catalog

The NHGRI-EBI Catalog of human genome-wide association studies



Examples: breast carcinoma, rs7329174, Yao, 2q37.1, HBS1L, 6:160000000-250000000

- Let's have a look at a specific variant, rs143383.

Search results for *rs143383*

V rs143383

Location: 20:35438203 **Cytogenetic region:** 20q11.22 **Most severe consequence:** 5 prime utr variant **Mapped gene(s):** GDF5

Associations **6** Studies **6**

G GDF5

Description: growth differentiation factor 5

Location: 20:35433347-35454746 **Cytogenetic region:** 20q11.22 **Biotype:** protein coding

Associations **92** Studies **67**

Human Traits & Diseases Databases

GWAS catalog

V rs143383

Location: 20:35438203 **Cytogenetic region:**20q11.22 **Most severe consequence:** 5 prime utr variant **Mapped gene(s):** GDF5

Associations **10** Studies **10**

Available data: Associations **10** Studies **10** Traits **5** Linkage disequilibrium (LD)

Download Associations 

Associations **10**

Show 5 entries

Column visibility Export Clear search

Variant and risk allele	P-value	P-value annotation	RAF	OR	Beta	CI	Mapped gene	Reported trait	Trait(s)	Background trait(s)	Study accession	Location
rs143383-G	2×10^{-16}	-	0.37	-	8.25 z score increase	-	GDF5	Vertex-wise cortical surface area	cortical surface area measurement	-	GCST90095130	20:35438203
rs143383-T	5×10^{-7}	-	NR	1.0495905	-	[1.03-1.07]	GDF5	Knee osteoarthritis	osteoarthritis, knee	-	GCST007090	20:35438203
rs143383-G	1×10^{-17}	-	0.3608	-	-	-	GDF5	Cortical surface area	cortical surface area measurement	-	GCST90091060	20:35438203
rs143383-T	8×10^{-18}	-	0.64	1.09287	-	[NR]	GDF5	Knee osteoarthritis	osteoarthritis, knee	-	GCST006925	20:35438203
rs143383-T	9×10^{-78}	-	NR	-	0.06367727 unit increase	[0.057-0.07]	GDF5	Height	body height	-	GCST008839	20:35438203

Showing 1 to 5 of 10 entries

« 1 2 »

Human Traits & Diseases Databases

GWAS catalog

GDF5

Description: growth differentiation factor 5

Location: 20:35433347-35454746 **Cytogenetic region:** 20q11.22 **Biotype:** protein coding

Associations **131** Studies **98**



Human Traits & Diseases Databases GWAS catalog



GWAS Catalog

The NHGRI-EBI Catalog of human genome-wide association studies

Search the catalog

Examples: Parkinson disease, rs3093017, Yao, 2q37.2, HBS1L, 6:167120000-167130000, GCST90132222, PMID:35241825



Are you accessing the GWAS catalog programmatically? The newly designed REST API v2.0 is here! Find out more in our [blog post](#).

Download

Download a full copy of the GWAS Catalog in spreadsheet format as well as current and older versions of the GWAS diagram in SVG format.

Summary statistics

Documentation and access to full summary statistics for GWAS Catalog studies where available.

Submit

Submit summary statistics to GWAS Catalog.

Learn

Including FAQs, our curation process, data sources, training materials, related resources, a list of abbreviations and API documentation.

Diagram

Explore an interactive visualisation of all SNP-trait associations with genome-wide significance ($p \leq 5 \times 10^{-8}$).

Population descriptors

An introduction to our data extraction and standardisation process.

Documentation

Summary statistics eligibility, submission information, standards and harmonisation.

API

A guide to how to access summary statistics data programmatically.

Available studies

A list of all studies for which summary statistics are available in the Catalog.

Studies with available summary statistics

Users can access summary statistics from the Catalog FTP site, which is updated nightly following submission. They can also be accessed in the tables below (separate tables for data associated with a journal publication "Published studies with summary statistics" and data shared before or without journal publication "Unpublished studies with summary statistics"). Metadata associated with summary statistics is available in the FTP table for each study and can also be downloaded in bulk from Catalog.

Files are in either raw form (summary statistics) or in a format that is suitable for the LDSC (LD score regression) analysis. The majority of data are made available either through LDSC or LDSC (LD score regression) analysis. For more information on individual studies, please visit "Single Studies" column.

List of published studies with summary statistics

Study	Published ID	Study accession	Pub. date	Journal	Title	Reported trait	Phenotype	Phenotype description	FTP Download
Maguire	2594000	GCST006477	2019-08-25	Nat Genet	Whole-genome analysis of genome-wide association studies	Depression	Major depressive disorder	FTP Download	
Maguire	2594000	GCST006475	2019-08-25	Nat Genet	Whole-genome analysis of genome-wide association studies	Depressed affect	Major depressive disorder	FTP Download	
Maguire	2594000	GCST006476	2019-08-25	Nat Genet	Whole-genome analysis of genome-wide association studies	Worry	Major depressive disorder	FTP Download	
Maguire	2594000	GCST006475	2019-08-25	Nat Genet	Whole-genome analysis of genome-wide association studies	Neuroticism	Major depressive disorder	FTP Download	
Collier-Land	8027403	GCST006490	2019-08-25	Commun Biol	Genome-wide association study for risk-taking	Self-reported risk-taking behaviour	Risk-taking behaviour	FTP Download	

Index of [pub/databases/gwas/summary_statistics/GCST006001-GCST007000/GCST006477](#)

Name	Last modified	Size	Description
Parent Directory		-	
harmonised/	2025-02-04 12:17	-	
readme_ctg.txt	2018-11-26 14:34	2.4K	
sumstats_depression_ctg_format.txt.gz	2018-11-26 14:34	298M	

Human Traits & Diseases Databases ...

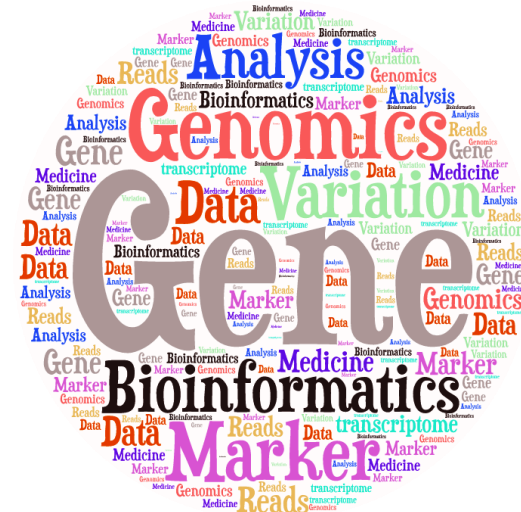
Exercise 3

1. Try to find more information about rs2075650 and the gene that is located in GWAS catalogue and OMIM databases.



Publicly available bioinformatics resources

- Human Genome Browsers
- Nucleotide Sequence Databases
- Protein Sequence Databases
- Database Searching by Sequence Similarity
- Protein Domains: Databases and Search Tools
- Human Traits & Diseases Databases
- Phylogeny & Taxonomy
- **Databases of other Organisms**
- Gene Prediction
- Gene Expression Databases
- Gene Regulation
- Metabolic, Gene Regulatory & Signal Transduction Network Databases
- Publications Database



Databases of other Organisms

- Mouse: http://www.ensembl.org/Mus_musculus/Info/Index

Ensembl BLAST/BLAT | VEP | Tools | BioMart | Downloads | Help & Docs | Blog

Mouse (GRCm38 p6) Location: 2 155,941,023-155,945,367 Gene: Gdf5 Transcript: Gdf5-201 Jobs

Gene-based displays

- Summary
 - Splice variants
 - Transcript comparison
 - Gene alleles
- Sequence
 - Secondary Structure
- Comparative Genomics
 - Genomic alignments
 - Gene tree
 - Gene gain/loss tree
 - Orthologues
 - Paralogues
 - Ensembl protein families
- Strains
- Ontologies
 - GO: Molecular function
 - GO: Biological process
 - GO: Cellular component
- Phenotypes
- Genetic Variation
 - Variant table
 - Variant image
 - Structural variants
- Gene expression
 - Pathway
 - Regulation
- External references
 - Supporting evidence
 - History
 - Gene history

Gene: Gdf5 ENSMUSG00000038259

Description growth differentiation factor 5 [Source: MGI Symbol; Acc: MGI:95588] [G](#)

Gene Synonyms CDMP-1, bp, brp, cartilage-derived morphogenetic protein-1

Location Chromosome 2: 155,941,023-155,945,367 reverse strand. GRCm38: CH000996.2

About this gene This gene has 1 transcript ([splice variant](#)), 280 orthologues, 32 paralogues, is a member of 1 Ensembl protein family and is associated with 48 phenotypes

Transcripts [Hide transcript table](#)

Show/Hide columns (1 hidden) [Filter](#)

Name	Transcript ID	bp	Protein	Biotype	CCDS	UniProt	Flags
Gdf5-201	ENSMUST00000040162.2	2320	496aa	Protein coding	CCDS16958.6	P43027.6	TSL:1; GENCODE basic; APPRIS P1

Summary

Name [Gdf5](#) (MGI Symbol)

CCDS This gene is similar to a CCDS gene on Mouse GRCm38: [CCDS16958.1](#) [G](#)

UniProtKB This gene has proteins that correspond to the following UniProtKB identifiers: [P43027](#) [G](#)

Ensembl version ENSMUSG00000038259.4

Gene type Protein coding

Annotation for this gene includes both automatic annotation from Ensembl and Havana manual curation, see [article](#).

Alternative genes This gene corresponds to the following database identifiers:
Havana gene: [OTTMUSG00000015869](#) [G](#)

[Go to Region in Detail for more tracks and navigation options \(e.g. zooming\)](#)

Regulation Legend CTCF Open Chromatin Promoter Promoter Flank

Gene Legend Protein Coding merged Ensembl/Havana

Databases of other Organisms

- Mouse: <http://www.informatics.jax.org/>

MGI Mouse Genome Informatics

Search Download More Resources Submit Data Find Mice (IMSR) Analysis Tools Contact Us Browsers

Keywords, Symbols, or IDs (exact phrase) Quick Search

Type your search here

Or use topic specific search and analysis tools:

- Genes
- Phenotypes & Mutant Alleles
- Human-Mouse: Disease Connection
- Gene Expression Database (GXD)
- Recombinase (cre)
- Function
- Strains, SNPs & Polymorphisms
- Vertebrate Homology
- Mouse Models of Human Cancer
- Batch Data and Analysis Tools
- Nomenclature

Getting Started:

- Introduction to mouse genetics
- How to use MGI (Text & Video)
- Cre Portal Tutorial

Community Interest

MGI Workshops

- If you would like to host an MGI workshop at your institution, [contact User Support](#) to discuss the details.
- A typical MGI workshop includes a brief presentation ([slides](#)) and a hands-on, interactive tutorial ([tutorial](#)).
- If you would like to schedule a remote interactive session to learn to use MGI, [contact User Support](#) to discuss the details.

Research Highlights

- Improving laboratory animal genetic reporting: LAG-R guidelines. [Nat Commun](#). 2024. 2024 Jul 2;15(1):5574.
- Into the Wild: A novel wild-derived inbred strain resource expands the genomic and phenotypic diversity of laboratory mouse models. [PLoS Genetics](#). 2024. Apr 10;20(4):e1011228.
- Resolution of structural variation in diverse mouse genomes reveals chromatin remodeling due to transposable elements. [Cell Genom](#). 2023 Apr 5;3(5):100291.

Other Resources

[Mouse Resources for COVID-19 Research](#): The information in this resource summarizes information about mouse strains and genes relevant to the study of coronavirus infection and pathology.

The [International Mouse Phenotyping Consortium](#) project is systematically phenotyping knockout mice from the mutant ES cells produced by the International Mouse Knockout Consortium. Data will be integrated into MGI as available.

MGI is the international database resource for the laboratory mouse, providing integrated genetic, genomic, and biological data to facilitate the study of human health and disease.

About Us MGI Publications Cite Us

Now with more SNPs
Including Sanger MGP SNP data

What's new at MGI updated July 26, 2025

- New Cell Ontology Browser and Filtering by Cell Type. [Read more.](#)
- RNA-seq experiment type querying now available in the RNA-Seq and Microarray Experiment Search. [Read more.](#)
- MGI SNP Function Classes are back and new Regulates Expression Relationships are now displayed on Gene pages. [Read more.](#)
- Ensembl release 113 regulatory annotation has been incorporated into MGI. [Read more.](#)
- Expression Profile Search redesigned and enhanced. [Read more.](#)
- SNP data increased 4-fold. [Read more.](#)
- MGI 1.0 was released on the World Wide Web 30 years ago. [Read more.](#)

MGI Statistics More MGI news

Databases of other Organisms

- Mouse: <https://www.mousephenotype.org/>

IMPC
International Mouse Phenotyping Consortium

ABOUT THE IMPC DATA HUMAN DISEASES PUBLICATIONS NEWS BLOG

IMPC Survey - Please provide your input!
The IMPC is at a funding crossroads and seeking community input. Help shape the future of IMPC by taking part in our [short survey](#).

Data release 23 is now here with over 100 million data points! Total of 9,277 knockout genes are phenotyped. Read more [here](#).

GENES PHENOTYPES HELP, NEWS, BLOG

Search All 9277 Knockout Data...

What you need to know about IMPC data

What you need to know about IMPC data

- » We generate our own data, it is **not aggregated** from publications
- » We are targeting genes that are yet to be studied
- » We have **standardised** production methods to knock out genes
- » Mutants and wildtype mice tested under the **same phenotyping protocols**

[Learn more about IMPC data](#)

9277 total knockout genes phenotyped

- » Number of phenotyped lines: 9994
- » Statistically Significant Calls: 113803
- » Data Release Version: 23
- » Published: 23 April 2025

[View Release Full Report](#)

Databases of other Organisms

- Zebrafish: <http://zfin.org/>

The screenshot shows the ZFIN website with a teal header containing navigation links: Research, Genomics, Resources, Community, and Support. A 'Sign In' link is on the right. Below the header is a large banner image of a zebrafish embryo with glowing spots. Overlaid on the banner is a search bar with the text 'Search expert curated zebrafish data'. Below the search bar is a dropdown menu set to 'Any' and a text input field containing 'heart contraction abnormal'. A 'Search' button is to the right of the input field. Below the banner, there are several white boxes with teal borders and arrows pointing right, each representing a different data category: Genes, Expression, About ZFIN, Mutants/Tg, Antibodies, BLAST, Publications, and Additional Resources. The 'About ZFIN' box contains a paragraph describing the database and a 'Learn More' link. The 'Additional Resources' box contains links to 'Data Mining', 'ZebrafishMine', and 'BioMart'.

Search expert curated zebrafish data

Any heart contraction abnormal Search

Fig. 1 of Monroe et al., 2016

Genes
Search for genes, transcripts, clones, and other markers

Expression
Search for gene expression data, and annotated images

About ZFIN
The Zebrafish Information Network (ZFIN) is the database of genetic and genomic data for the zebrafish (*Danio rerio*) as a model organism. ZFIN provides a wide array of expertly curated, organized and cross-referenced zebrafish research data.
[Learn More](#)

Mutants/Tg
Search for mutants, knockdowns, transgenics, and affected phenotypes

Antibodies
Search for antibodies by gene, labeled anatomy, and other attributes

BLAST
Align nucleotide and protein sequences with zebrafish datasets

Publications
Search for zebrafish research publications and scientific literature

Additional Resources
Data Mining
[ZebrafishMine](#) [BioMart](#)

Databases of other Organisms

- Drosophila: <http://flybase.org/>

The screenshot shows the FlyBase website, a database for *Drosophila* genes and genomes. The header includes the FlyBase logo, a release date of Oct 14, 2020, and a navigation bar with links like Home, Tools, Downloads, Links, Community, Species, About, Help, and Archives. A search bar with 'J2G' and a 'Jump to Gene' button is also present. Below the navigation bar are several featured tools and resources, including BLAST, GBrowse, JBrowse, Antibodies, CRISPR, RNAi, GO Phenotype, Anatomy, Disease, and more. A 'QuickSearch' section is visible, with tabs for Human Disease, Protein Domains, Gene Groups, Pathways, GO, and Data Class. A search bar with a dropdown menu and a 'Search' button is provided. Below the search bar, there are icons for various biological systems: Adipose, Circulatory, Excretory, Muscle, Imaginal Precursor, Integumentary, Tracheal, Digestive, Nervous, and Reproductive. The bottom of the page features a large red banner with the Helmholtz Zentrum München logo and the text 'German Research Center for Environmental Health'.

FB2020_05, released Oct 14, 2020
A Database of *Drosophila* Genes & Genomes

Home Tools Downloads Links Community Species About Help Archives J2G Jump to Gene Go

D. melanogaster D. virilis A. mellifera BLAST
GO+ GBrowse JBrowse
Antibodies Interactions RNAi CRISPR Resources
ON OFF RNA-Seq
GO PHENOTYPE ANATOMY DISEASE MORE Vocabularies
ImageBrowse
FIELD DATA X M L Sequence Batch Download

Using JBrowse on FlyBase

QuickSearch
Human Disease Protein Domains Gene Groups Pathways GO Data Class
Search FlyBase Homologs GAL4 etc Expression Phenotype References
Everything Search
Click here to submit multiple IDs/symbols.
Note: Wild cards (*) can be added to your search term

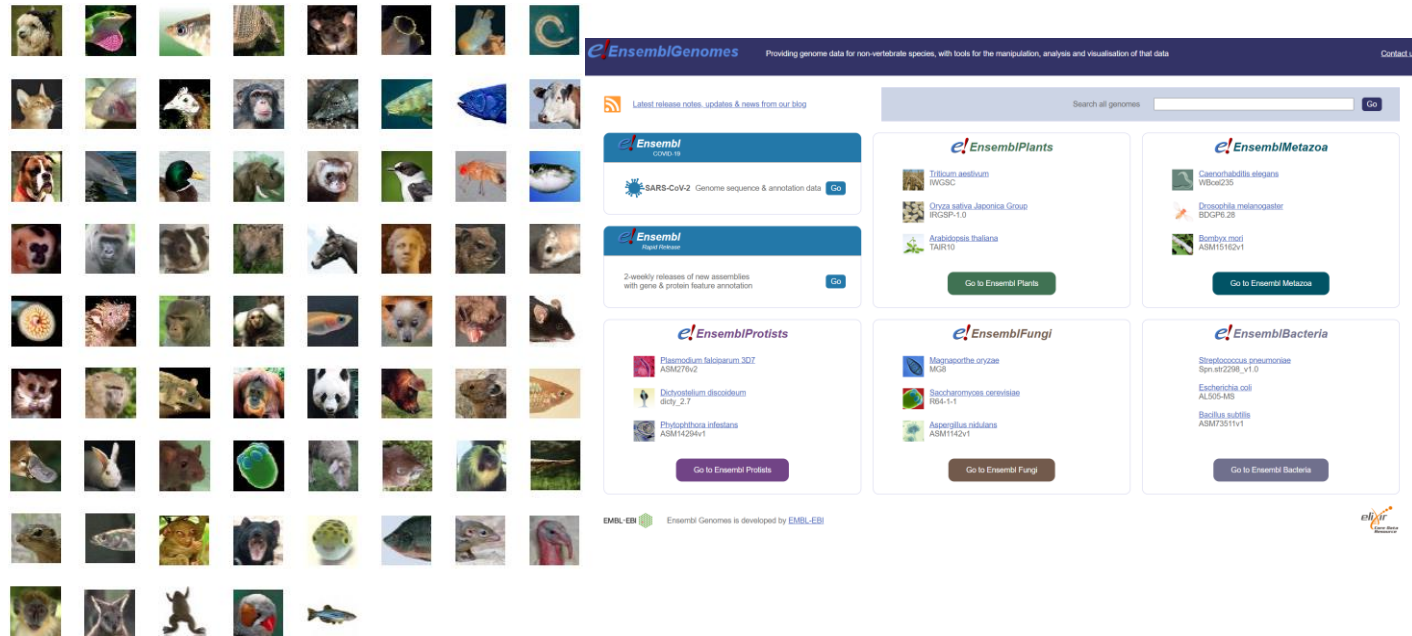
FAST-TRACK YOUR PAPER FLYBASE NEWS FLY BOARD COMMUNITY NEWS
MEETINGS COURSES FLYBOOK

Resources for:
PUBLIC TEACHERS STUDENTS BIOMEDICAL RESEARCH POSITIONS AVAILABLE

Adipose Circulatory Excretory Muscle Imaginal Precursor
Integumentary Tracheal Digestive Nervous Reproductive

Databases of other Organisms

- Ensembl : >300 vertebrate species (<https://www.ensembl.org/info/about/species.html>) & >45000 genomes from non-vertebrate species (<http://ensemblgenomes.org/>)



The screenshot displays the Ensembl Genomes website interface. On the left, a grid of 48 small images shows a variety of animals, including mammals, birds, reptiles, and invertebrates. The main content area features the Ensembl Genomes logo and tagline: "Providing genome data for non-vertebrate species, with tools for the manipulation, analysis and visualisation of that data". Below this, there are several sections:

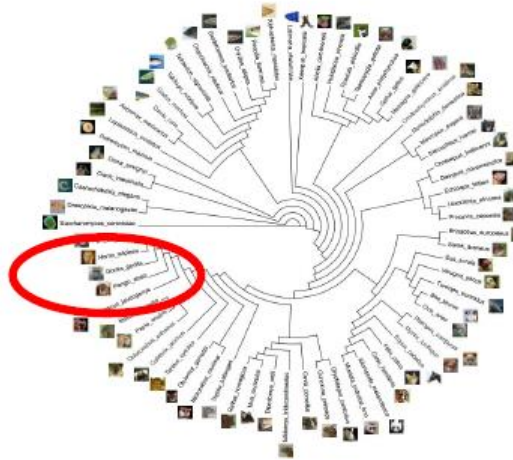
- Latest release notes, updates & news from our blog**: A link to the latest release notes.
- Ensembl COVID-19**: A section for SARS-CoV-2 genome sequence and annotation data.
- Ensembl Plant Release**: A section for 2-weekly releases of new assemblies with gene & protein feature annotation.
- Ensembl Plants**: A section listing plant genomes, including *Triticum aestivum* (IWGSC), *Oryza sativa* (Japonica Group) (IRGSP-1.0), and *Arabidopsis thaliana* (TAIR10).
- Ensembl Fungi**: A section listing fungal genomes, including *Magnaporthe oryzae* (MG9), *Saccharomyces cerevisiae* (RM1-1), and *Aspergillus nidulans* (ASMI142v1).
- Ensembl Metazoa**: A section listing metazoan genomes, including *Caenorhabditis elegans* (WBC235), *Drosophila melanogaster* (BDGP6.28), and *Bombus morio* (ASMH15162v1).
- Ensembl Bacteria**: A section listing bacterial genomes, including *Streptococcus pneumoniae* (GPM2208, v1.0), *Escherichia coli* (AL505-MB), and *Bacillus subtilis* (ASMB7351v1).

At the bottom of the page, there is a footer with the Ensembl logo and the text "Ensembl Genomes is developed by EMBL-EBI".

Databases of other Organisms ...

Exercise 4

- Explore rs2075650 in the databases of other organisms.
 - *Hint: you can search by rsID or (mapped) gene id or phenotype.*



- 



Gene Expression Databases

- Genotype-Tissue Expression (GTEx) : Correlations between genotype and tissue-specific gene expression (<https://www.gtexportal.org/home/>).

The Genotype-Tissue Expression (GTEx) Portal is a comprehensive public resource for researchers studying tissue and cell-specific gene expression and regulation across individuals, development, and species, with data from 3 NIH projects.

GTEx
The Adult GTEx project is a comprehensive resource of WGS, RNA-Seq, and QTL data from samples collected from 54 non-diseased tissue sites across ~1000 adult individuals.
[Explore >>](#)

dGTEx
The Developmental GTEx (dGTEx) project is a new effort to study development-specific genetic effects on gene expression and to establish a new data analysis and tissue biobank resource.
*Data Not Yet Available [Explore >>](#)

NHP-dGTEx
The Non-Human Primate Developmental GTEx (NHP-dGTEx) project is a complement to dGTEx in 2 translational non-human primate model species: the rhesus macaque and common marmoset.
*Data Not Yet Available [Explore >>](#)

Explore the Adult GTEx data
dGTEx and NHP-dGTEx will be available at a later time.

[Browse](#)
By Tissue

Filter and search all data by tissue

Announcements
2025-01-15 Developmental GTEx paper published in Nature
Our new perspective article describing the human and non-human primate...
2025-01-15 Register for the Mutations in Time and Space conference
The Mutations in Time and Space conference will be held...

Gene Expression Databases

- Expression Atlas: Differential and Baseline Expression (<http://www.ebi.ac.uk/gxa/home> or through Ensembl).

The screenshot shows the Expression Atlas website. The header features the logo and the title "Expression Atlas" with the subtitle "Gene expression across species and biological conditions". Navigation links include Home, Browse experiments, Download, Release notes, FAQ, Help, Licence, About, and Support. A search bar is present with the text "Search across 67 species, 4,562 studies, 161,199 assays". Below the search bar, there are input fields for "Gene / Gene properties", "Species" (set to "Any"), and "Biological conditions". A "Search" button is located below these fields. The main content area displays a grid of species cards, each with an icon, the species name, the number of experiments, and the number of baseline and differential experiments. The species shown are Homo sapiens, Mus musculus, Rattus norvegicus, Drosophila melanogaster, Sus scrofa, and Gallus gallus. A "Show experiments of all species" button is at the bottom of the grid.

Species	Experiments	Baseline	Differential
Homo sapiens	1651	123	1528
Mus musculus	1304	65	1239
Rattus norvegicus	186	14	172
Drosophila melanogaster	150	5	145
Sus scrofa	47	16	31
Gallus gallus	39	4	35

- [illegible]

Gene Expression Databases ...

Exercise 5

- Does rs2075650 affect the expression of the gene?
 - Hint: Use GTEx portal
- In which tissue *TOMM40* is expressed? Any connections with Alzheimer disease?
 - Hint: Use Expression Atlas



Publications Database

- Genome Browsers
- Nucleotide Sequence Databases
- Protein Sequence Databases
- Database Searching by Sequence Similarity
- Protein Domains: Databases and Search Tools
- Human Traits & Diseases Databases
- Phylogeny & Taxonomy
- Databases of other Organisms
- Gene Prediction
- Gene Expression Databases
- Gene Regulation
- Metabolic, Gene Regulatory & Signal Transduction Network Databases
- Publications Database

- PubMed (<https://www.ncbi.nlm.nih.gov/pubmed>) comprises over 34 million citations.
- How do I search PubMed?
 - Identify the key concepts for your search.
 - Enter the terms* in the search box.
 - Keywords
 - Author name
 - Journal name
 -
 - Suggestions will display as you type your search terms.
 - Click Search.



Publications Database

- Anatomy of the summary results:

- ☐ [Policy Issues in the Development and Adoption of Biomarkers for Molecularly Targeted Cancer Therapies: Workshop Summary.](#)
1. National Cancer Policy Forum, Board on Health Care Services, Institute of Medicine.
Washington (DC): National Academies Press (US); 2015.
PMID: 25855848 [Free Books & Documents](#)
[Similar articles](#)
- ☐ [Four-wave mixing experiments with extreme ultraviolet transient gratings.](#)
2. Bencivenna F, Cucini R, Capotondi F, Battistoni A, Mincigrucci R, Giangrisostomi E, Gessini A, Manfreda M, Nikolov IP, Pedersoli E, Principi E, Svetina C, Parisse P, Casolari F, Danailov MB, Kiskinova M, Masciovecchio C. Nature. 2015 Apr 9;520(7546):205-8. doi: 10.1038/nature14341.
PMID: 25855456
[Similar articles](#)
- ☐ [Molecular imaging of angiogenesis after myocardial infarction by \(111\)In-DTPA-cNGR and \(99m\)Tc-sestamibi dual-isotope myocardial SPECT.](#)
3. Hendrikx G, De Saint-Hubert M, Dijkgraaf I, Bauwens M, Douma K, Wiertz R, Pooters I, Van den Akker NM, Hackeng TM, Post MJ, Mottaghy FM. EJNMMI Res. 2015 Jan 28;5:2. doi: 10.1186/s13550-015-0081-7. eCollection 2015.
PMID: 25853008 [Free PMC Article](#)
[Similar articles](#)

title

authors

journal title
abbreviation

volume & issue

e-pagination

publication date



Publications Database ... Exercise 6

- Try to find more information about rs2075650 and the gene that is located.
- Can you find a paper for which:
 - “Lopes” is one of the authors
 - Dealing with “osteoarthritis”
 - The name of the journal is “Lancet”



Variant:

Ensembl:

- ✓ position,
- ✓ consequences (transcript and regulatory region)
- ✓ 1kG frequencies,
- ✓ GWAS signals around the variant,
- ✓ regulatory marks around the variant
- ✓ nearby genes.

ExAC: allele frequencies

GTEx: associated gene.

Pubmed: papers where the variant is cited.

Gene:

Ensembl:

- ✓ Position
- ✓ Associated phenotypes
- ✓ Gene ontology annotations

Uniprot:

- ✓ Function, Tissue, Localization, Disease

OMIM:

- ✓ gene annotation
- ✓ associated diseases

GWAS catalog:

- ✓ Associated variants and association details.

Gene Expression Atlas:

- ✓ Tissues in which expression of the gene was observed

GTEx:

- ✓ List of variants that affects the expression of the gene.

Mouse Genomics Informatics:

- ✓ Mouse phenotype of the gene.



Dr Daniel Suveges
Senior Bioinformatician

Team144 pipeline

Variant details

General information:

rsID: [rs143384](#)
Chromosome:start-end: [20:34025756-34025756](#)
Allele string: A/G
MAF: 0.438898
Consequence: 5_prime_UTR_variant
Variation type: SNP
gerp (average gerp): 5.04 (3.497)
GWAVA score: 0.79
Ancestral allele: G
Minor allele: A
Synonyms: rs431839, rs61433024, rs765106, rs3748435, rs17422934
Phenotypes: [Height\(A\)](#),
(risk allele): [Infant length\(G\)](#),
Height(G)

GWAS signals

GWAS signals within 1Mbp of the variant:

rsID	SNPID	Distance	Trait	p-value	PMID
rs17310467	chr20:33545616	-481kbp	Hemostatic factors and hematological phenotypes	4E-34	22443383
rs11906160	chr20:33565755	-461kbp	Anticoagulant levels	1E-6	22216198
rs1535466	chr20:33718706	-308kbp	Height	1.8E-29	25282103
rs2295888	chr20:33722863	-303kbp	Prothrombin time	5E-13	22703881
rs6120849	chr20:33730387	-296kbp	Protein C levels	7E-37	20802025
rs6088735	chr20:33745676	-281kbp	Hemostatic factors and hematological phenotypes	4E-34	22443383
rs6060278	chr20:33753262	-273kbp	Hemostatic factors and hematological phenotypes	4E-34	22443383
rs867186	chr20:33764554	-262kbp	Anticoagulant levels	4E-9	22216198
rs867186	chr20:33764554	-262kbp	Coagulation factor levels	6E-37	20231535
rs867186	chr20:33764554	-262kbp	D-dimer levels	4E-6	21502573
rs867186	chr20:33764554	-262kbp	Hemostatic factors and hematological phenotypes	2E-6	22443383
rs867186	chr20:33764554	-262kbp	Protein C levels	1E-64	25376901
rs11167260	chr20:33775200	-251kbp	Amyotrophic lateral sclerosis	4E-6	24529757



Dr Daniel Suveges
Senior Bioinformatician

HelmholtzZentrum münchen

German Research Center for Environmental Health

Thank you!

HELMHOLTZ RESEARCH FOR
GRAND CHALLENGES

What is a genome assembly?

Sequence reads

```
CGGCCTTTGGGCTCCGCCTTCAGCTCAAGA
CAGCTGTCCCAGATGAC    ACTTAACTTCCCTCCCAGCTGTCC
GGGCTCCGCCTTCAGCTC  AACTTCCCTCCCAGCT    TCCCAGCTGTCCCAGATGACGCCATC
CGGCCTTTGGGCTCC    CAGATGACGCC    TCCGCCTTCAGCTCAAGACTTAACTTC
```

Match up overlaps

```
CGGCCTTTGGGCTCCGCCTTCAGCTCAAGA  AACTTCCCTCCCAGCT    CAGATGACGCC
                                TCCGCCTTCAGCTCAAGACTTAACTTC  TCCCAGCTGTCCCAGATGACGCCAT
                                GGGCTCCGCCTTCAGCTC    ACTTAACTTCCCTCCCAGCTGTCC
CGGCCTTTGGGCTCC                                CAGCTGTCCCAGATGAC
```

Genome assembly

```
CGGCCTTTGGGCTCCGCCTTCAGCTCAAGACTTAACTTCCCTCCCAGCTGTCCCAGATGACGCCAT
```

Ensembl ... the front page ... Exercise 1



1. Go to the species homepage for **Giant Panda**. What is the name of the genome assembly for Panda?
 - ASM200744v2, INSDC Assembly [GCA_002007445.2](#), Apr 2020
2. How long is the Panda genome (in bp)? How many coding genes have been annotated?
 - 2,444,060,653 & 20,857
3. Repeat for **Human**.
 - GRCh38.p13 (Genome Reference Consortium Human Build 38), INSDC Assembly GCA_000001405.28, Dec 2013
 - 3,099,750,718
 - 19,869 (excl 664 readthrough)



Ensembl ... Exploring variants ... Exercise 2

Explore rs2075650 (build 37).

1. What type of variants is? **SNP**
2. Where is this variant located (chr - position - strand)? **Chromosome 19:44892362 (forward strand)**
3. Are there big differences in allele frequencies between populations? **Not really. Max 8% in some sub-African pop and 10% in som AMR sub-pop too.**
4. What is the ancestral allele frequency in the Japanese in Tokyo (JPT) population from the 1000G set? **12%**
5. What is the least frequent genotype for this variant in the KHV population from the 1000G set? **G|G**
6. In how many transcripts is this variant found? **5**
7. Are all transcripts from the same gene? Yes, **TOMM40**
8. How many and which consequence type are predicted for this variant? **1, Intron**
9. How many phenotypes are associated with this variant? **67**
10. How many phenotypes are associated with the gene that this variant is located in? **1**
11. How many publications refer to rs2075650? **313**